ND

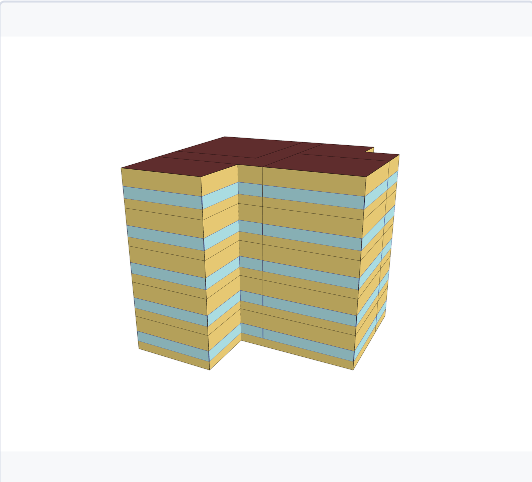
VERSION 2026-05 | DATE 2026-05-23

SECTION 1 — MODEL DESCRIPTION

BUILDING DESCRIPTION

Sector	Residential
Building Type	Multi-Unit
Building Sub Type	Apartment
size_of_build	MID Rise
Vintage	1945

3D MODEL



Three-dimensional geometric representation of the building model used for energy simulation and performance analysis.

BUILDING CHARACTERISTICS

Floor Area	1325.0 m²
Climate Zone	6A

HVAC SYSTEM

cooling system	Central Electric Heat Pump System
Heating system	Central Heat Pump / Boiler
Ventilation system	Corridor Exhaust + Fresh Air Make-Up
Typical COP cooling	3.5 – 5.0

ENVELOPE

INSULATION & AIR TIGHTNESS

Walls Ext [W/m2-K]	1.13
Walls Int [W/m2-K]	2.51
Roof [W/m2-K]	2.49
Slabs [W/m2-K]	1.14
Infiltration [m^3/h-m^2]	13.5

GLAZING

Glazing [W/m2-K]	3.32
SHGC	0.66

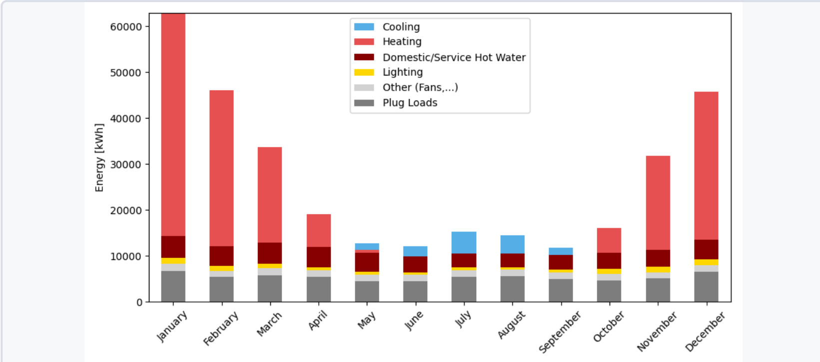
WWR

Average	30.0

ANNUAL END USES — ELECTRICITY [KWH]

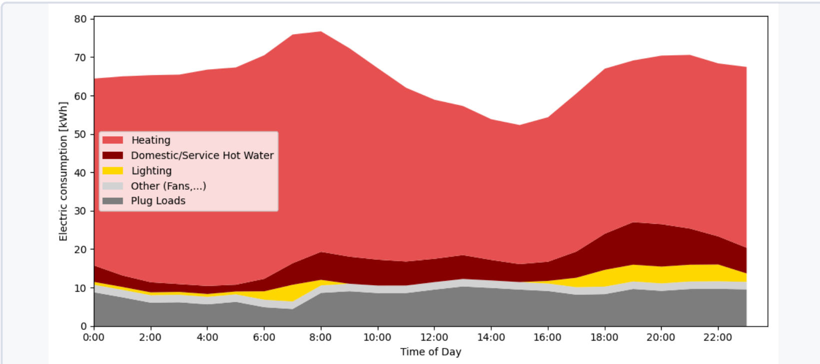
END USE	KWH/YR
Heating	169148.6
Cooling	14250.1
Interior Lighting	7086.2
Exterior Lighting	3183.4
Interior Equipment	110975.9
Fans	17069.6
Total End Uses	321713.7

MONTHLY CONSUMPTION BY END USE



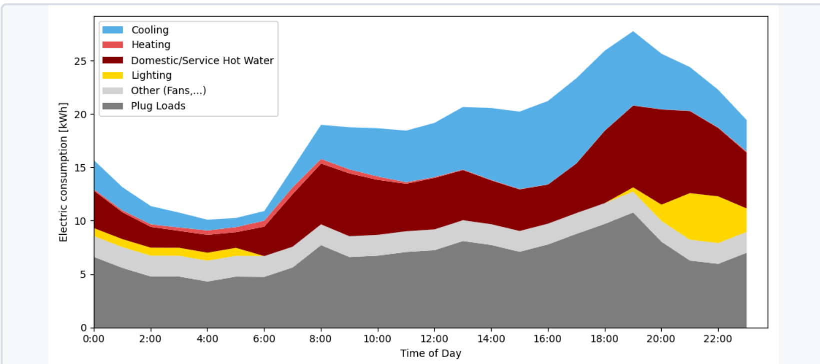
Monthly energy consumption breakdown by end use, showing the contribution of heating, cooling, lighting, domestic hot water, plug loads, and auxiliary systems throughout the year.

TYPICAL DAILY CONSUMPTION - WINTER



Stacked daily winter energy consumption profile showing the contribution of heating, lighting, domestic hot water, plug loads, and auxiliary systems throughout a typical winter day.

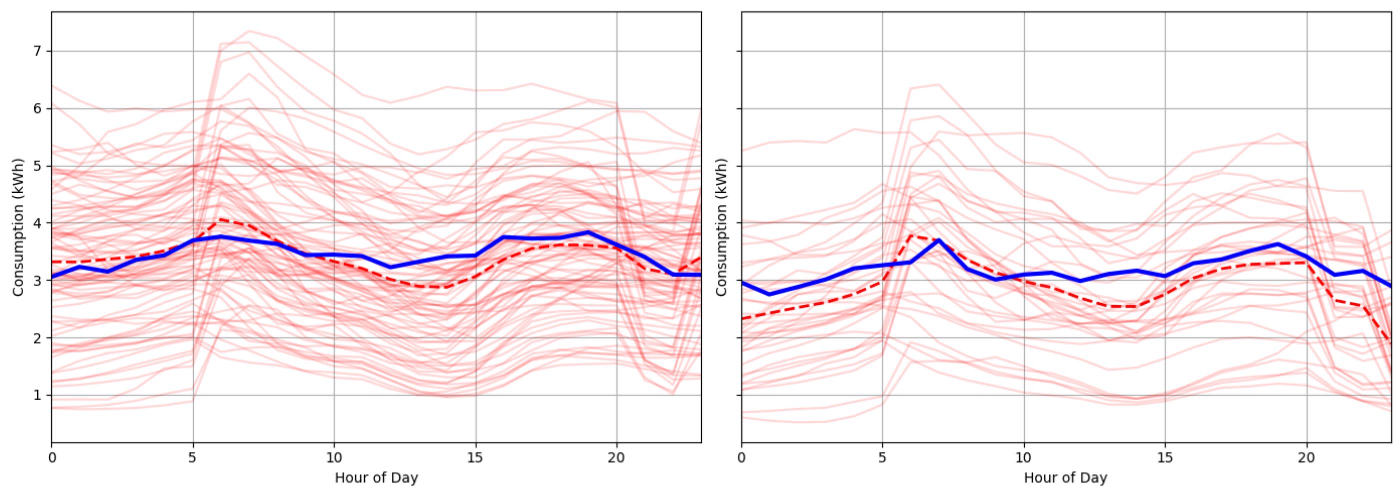
TYPICAL DAILY CONSUMPTION - SUMMER



Stacked daily summer energy consumption profile showing the contribution of cooling, lighting, domestic hot water, plug loads, and auxiliary systems throughout a typical summer day.

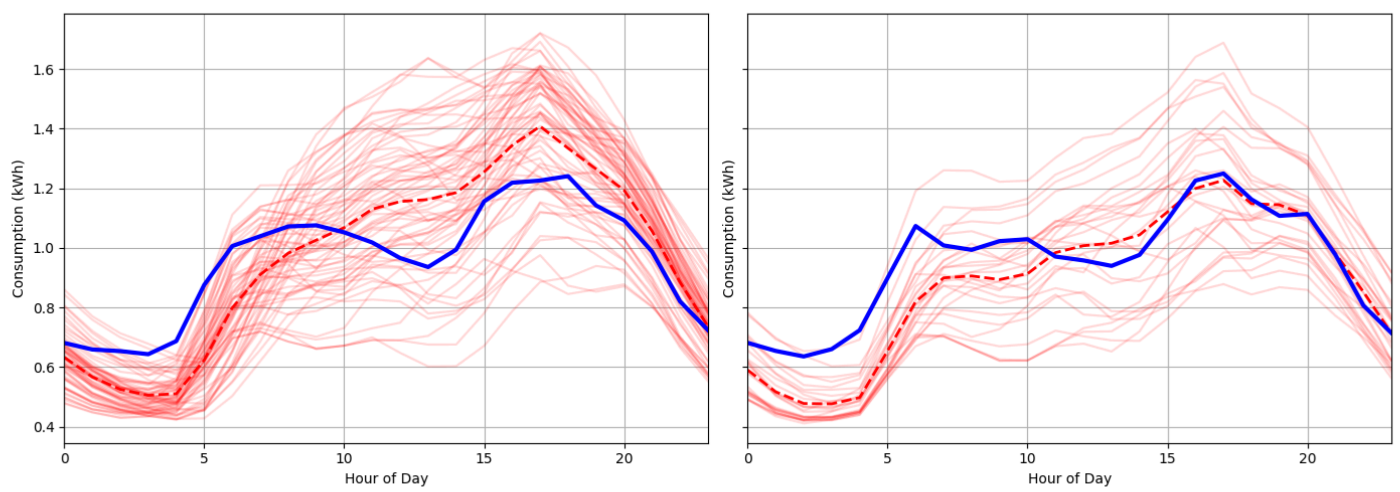
ENERGY USE INTENSITY	TOTAL CONSUMPTION	ELECTRICITY	NATURAL GAS
242.8 kWh/m²/yr	321711.0 kWh/yr	321714.0 kWh/yr	0 kWh/yr

TYPICAL DAY — WINTER



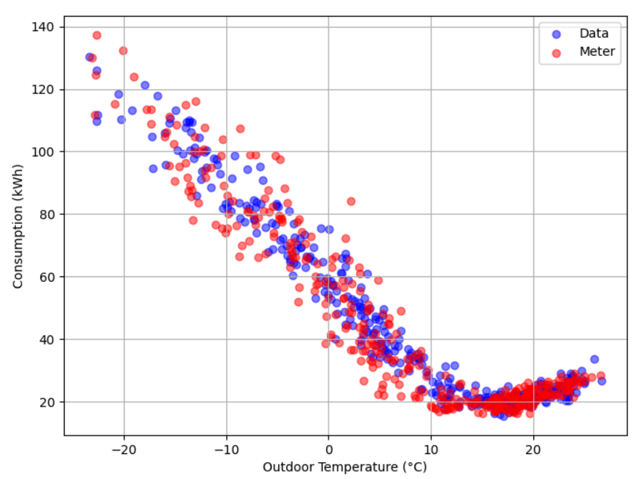
Winter daily consumption profile comparing simulated and measured hourly energy use. Red lines represent individual winter days from the model, the red dashed line indicates the average simulated profile, and the blue line shows measured data for comparison. The left plot represents weekdays, while the right plot represents weekends.

TYPICAL DAY — SUMMER



Summer daily consumption profile comparing simulated and measured hourly energy use. Red lines represent individual winter days from the model, the red dashed line indicates the average simulated profile, and the blue line shows measured data for comparison. The left plot represents weekdays, while the right plot represents weekends.

PRISM CURVE



PRISM curve comparing simulated (OPE) and measured energy consumption as a function of outdoor temperature, illustrating the relationship between weather conditions and building energy demand across the year.

COMPARISON TABLE — KEY INDICATORS

INDICATOR	MODEL	REFERENCE	DELTA
Yearly electricity consumption (kWh)	16086.08	16587.18	-501.1
Yearly baseload consumption (kWh/jour)	19.74	20.36	-0.6
Slope electricity heating (W/K)	-133.33	-127.08	-6.2
Slope climatisation cooling (W/K)	41.31	43.18	-1.9
Peak winter AM (kW)	7.34	6.77	0.6
Hour peak winter AM	7.0	6.0	1.0
Peak winter PM (kW)	6.42	6.55	-0.1
Hour peak winter PM	17.0	20.0	-3.0